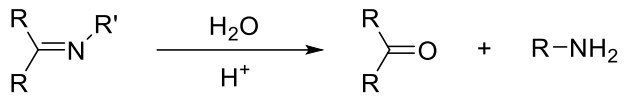
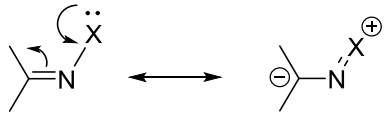
Imines are prone to hydrolysis.

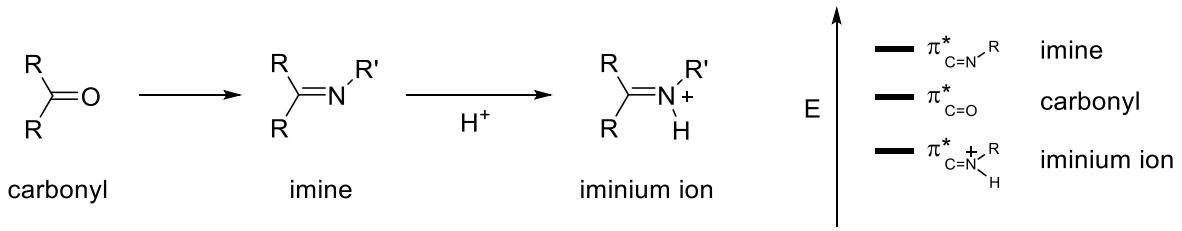


- The hydrolysis reaction is catalyzed by acid. The mechanism is exactly opposite.
- The first step is the protonation.
- Oximes and hydrazones are more stable due to the resonance that reduces the electrophilicity of the imine carbon. Stronger acid and heating conditions are required for the hydrolysis of oximes and hydrazones.

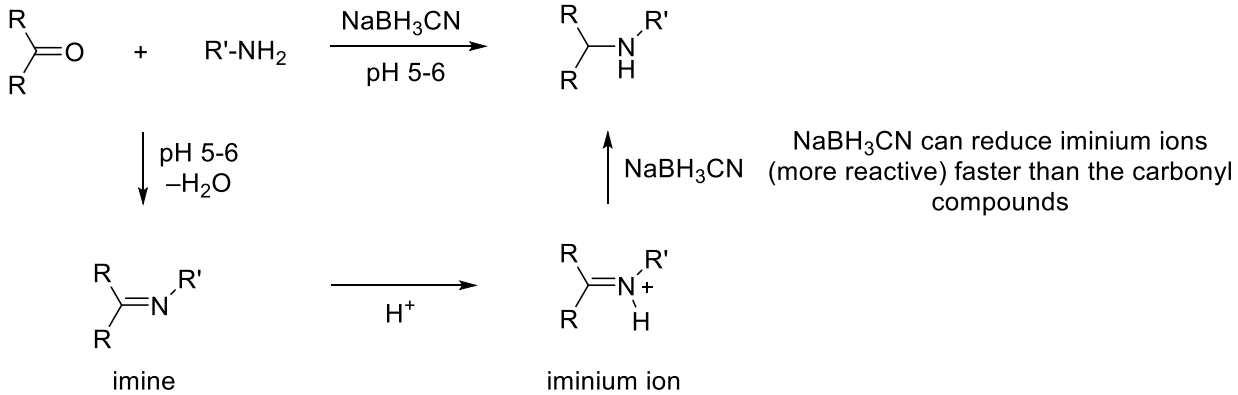
X = OH, NHR



Iminium ions are more electrophilic than the imines and the carbonyl compounds

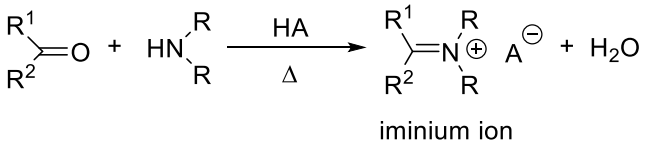


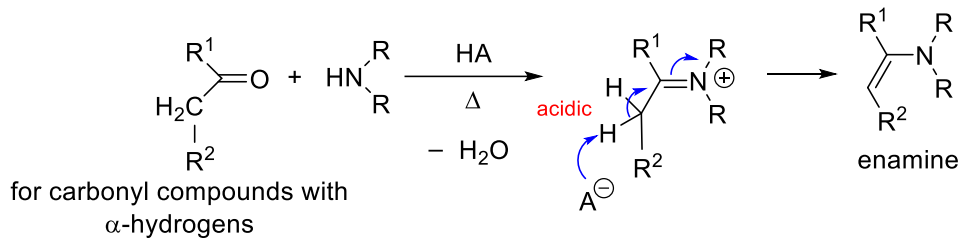
Reductive amination reaction



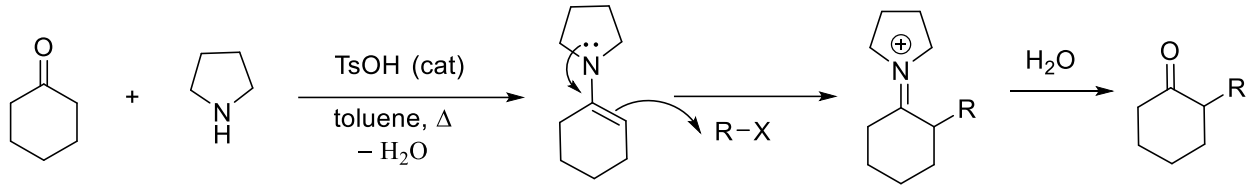
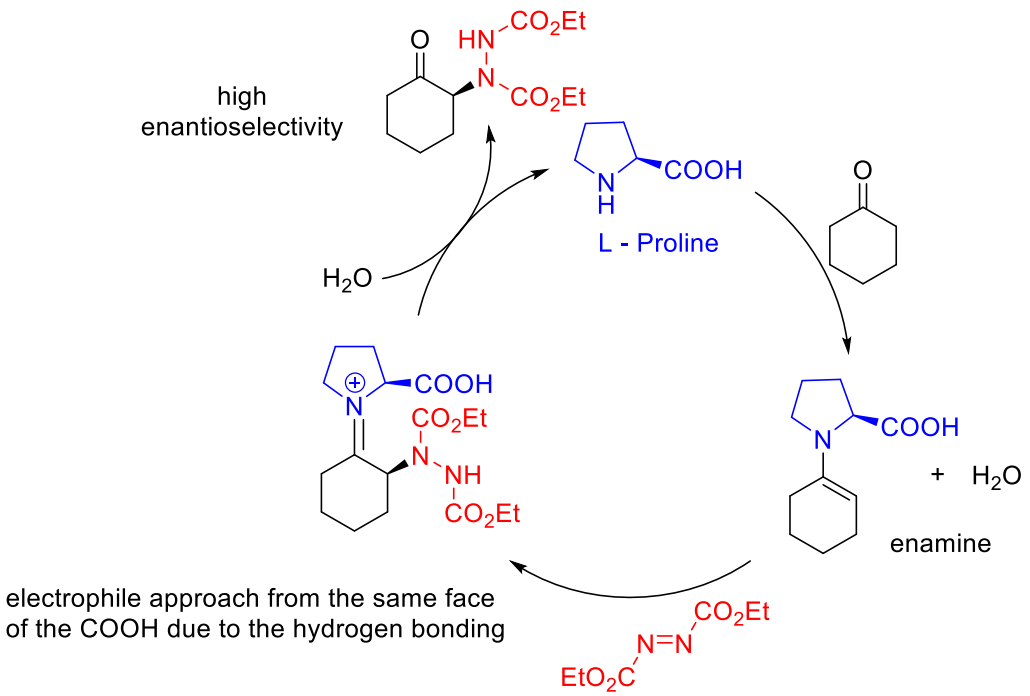
selective protonation of imine (pK_a ~ 10) over the carbonyl compounds (pK_a <0)

Reactions with secondary amines (formation of enamines)





Enamines are nucleophiles



Enamine activation (organocatalysis)